

UNIT - II

IIR - digital filter design

Analog filter

Digital filter

i) structure ✓
Resistor
L
C

i) summer, multiplier, delay element

ii) LT ✓
 $H(s) = \frac{Y(s)}{X(s)}$
analog Tr-fn ✓

ii) ZT ✓ → digital Tr-fn
 $H(z) = \frac{Y(z)}{X(z)}$

iii) $h(t)$ → filter coeff

iii) $h(n)$

iv) flexibility - less

iv) flexibility → More

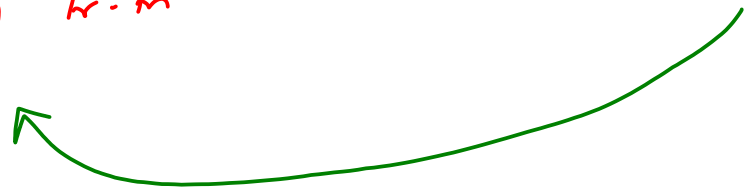
$H(s)$

$H(z)$

Analog Tr-fn

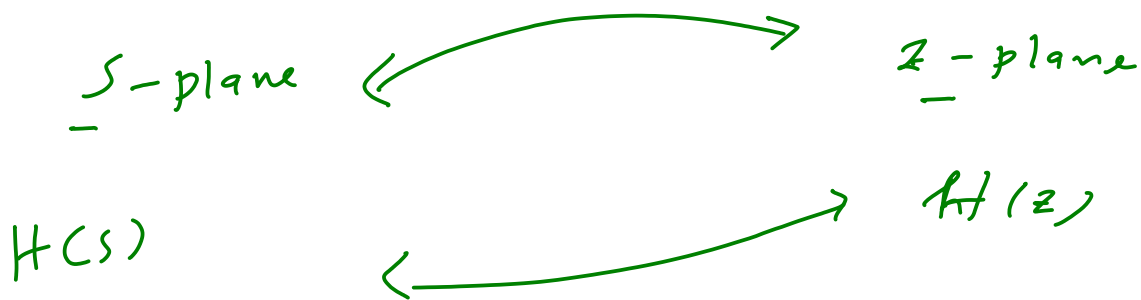
Mapping

Digital Tr-fn



Design of filter
analog
→ $H(s)$
→ $h(t)$ ✓

Design of digital filter
→ $H(z)$
→ $h(n)$



Relation between LT & ZT

LT $\rightarrow$ cont. signal $x(t)$

$$\text{LT}[x(t)] = x(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \quad \text{--- (1)}$$

$$\text{LT}[x(nT)] = x(s) = \sum_{n=-\infty}^{\infty} x(nT) e^{-snT} \quad \text{--- (2)}$$

ZT $\rightarrow$ digital signal $\rightarrow x(n)$

$$\text{ZT}[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \quad \text{--- (3)}$$

$$\text{eqn (2)} \Rightarrow x(s) = \sum_{n=-\infty}^{\infty} x(nT) \left(e^{sT}\right)^{-n} \quad \text{--- (4)}$$

compare (3) & (4)

$$x(s) = x(z) \quad \text{at} \quad \boxed{z = e^{sT}}$$

$z \rightarrow$ complex variable

$$z = r \cdot e^{j\omega}$$

(5)

$s \rightarrow$ complex variable

$$s = \sigma + j\omega$$

(1)

We know that

$$z = e^{sT}$$

$$\underline{r} \cdot e^{j\omega} = e^{(\sigma + j\omega)T} = \underline{e^{\sigma T}} \cdot e^{j\omega T} \quad (7)$$

$$\gamma = e^{\sigma T}$$

$$e^{j\omega} = e^{j\omega T}$$

$$\omega = \omega T$$

Analogy freq

Sampling period

Digital freq

$$\gamma = e^{\sigma T}$$

Mapping procedure between
 s -plane with z -plane

$$\underline{\sigma < 0} \rightarrow \sigma = -\infty$$

$$\gamma = e^{\sigma T} = e^{-\infty \cdot T} = 0$$

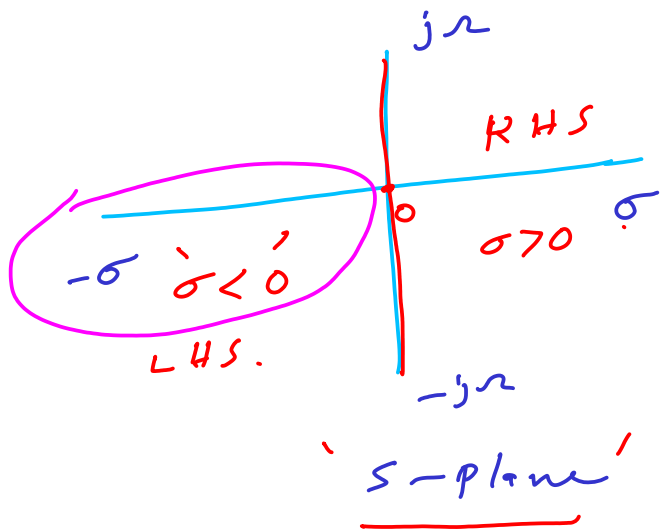
$$0$$

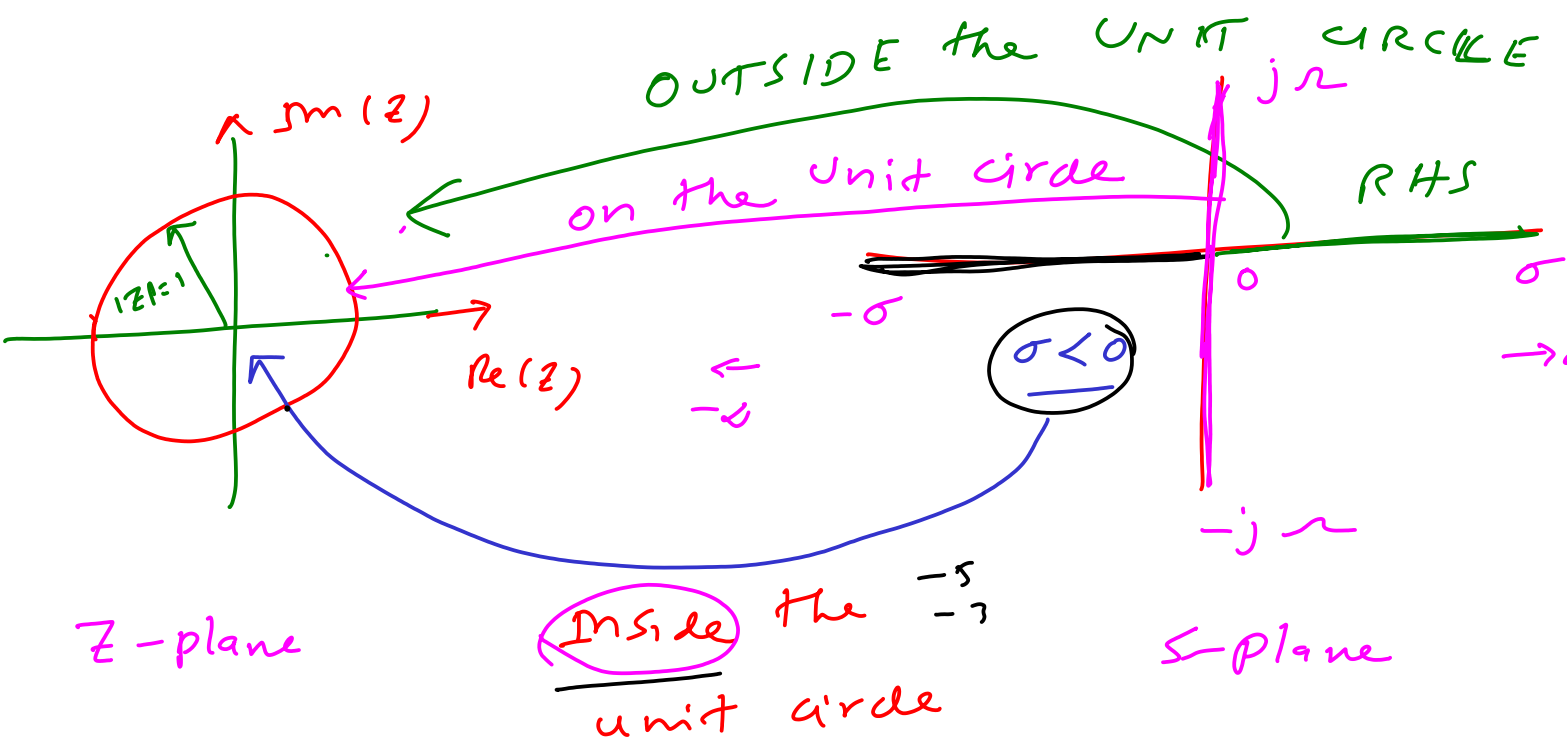
$$\underline{\sigma > 0} \rightarrow \sigma = \infty$$

$$\gamma = e^{\sigma T} = e^{\infty \cdot T} = \infty$$

$$\underline{\sigma = 0}$$

$$\gamma = e^{\sigma T} = e^0 = 1$$





s -plane $\longleftrightarrow$ z -plane
 LHS $\longrightarrow$ inside the unit circle
 RHS $\longrightarrow$ OUTSIDE the unit circle
 Imaginary axis $\longrightarrow$ on the unit circle

Design of Analog filter

" $H(s)$ " $\xrightarrow{\text{Stability}}$ $= \frac{Y(s)}{X(s)}$

'poles' $\rightarrow$ location $\leftarrow$ Stability $\rightarrow$ 'all poles' must lie in the LHS- s plane

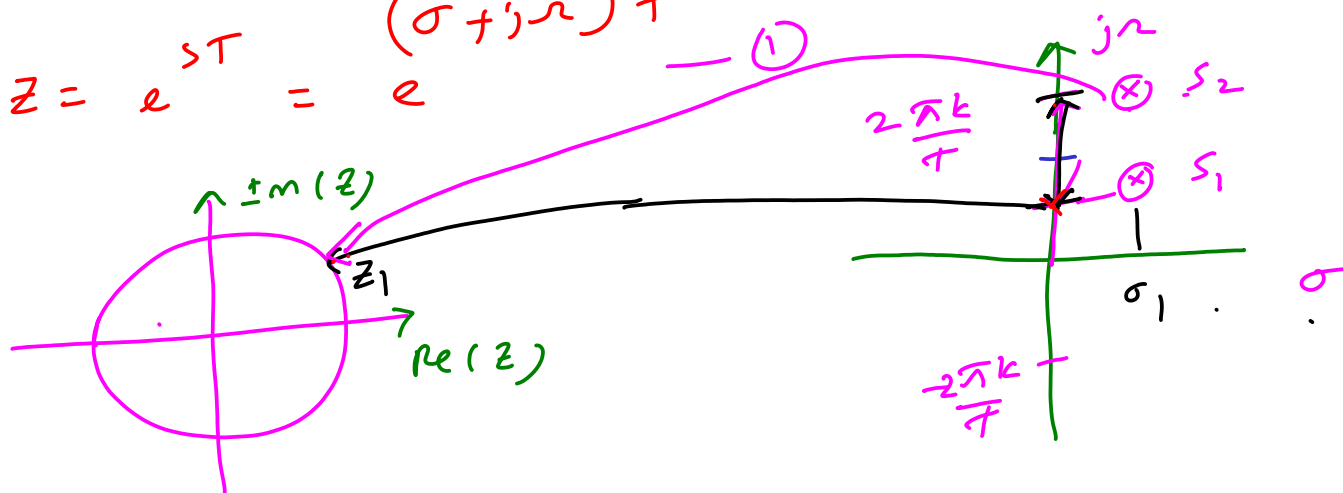
Digital filter $\rightarrow$ all poles $\rightarrow$ Must lie inside the unit circle

②

$$z = \sigma \cdot e^{j\omega}$$

$$s = \sigma + j\omega$$

$$z = e^{sT} = e^{(\sigma + j\omega)T}$$



$$s = s_1 = \sigma_1 + j\omega_1 \checkmark$$

$$z_1 = e^{s_1 T} = e^{(\sigma_1 + j\omega_1)T} = e^{\sigma_1 T} \cdot e^{j\omega_1 T} \quad \text{--- (2) } \checkmark$$

$$s = s_2 = \sigma_1 + j\left(\omega_1 + \frac{2\pi k}{T}\right) \checkmark$$

$$z_2 = e^{s_2 T} = e^{[\sigma_1 + j\omega_1 + j\frac{2\pi k}{T}]T}$$

$$= e^{\sigma_1 T} \cdot e^{j\omega_1 T} \cdot e^{j\frac{2\pi k}{T} T} = e^{\sigma_1 T} \cdot e^{j\omega_1 T} \cdot e^{j2\pi k}$$

$$z_2 = e^{\sigma_1 T} \cdot e^{j\omega_1 T} \quad \text{--- (3) } \checkmark$$

$$\omega \rightarrow \frac{2\pi k}{T}$$

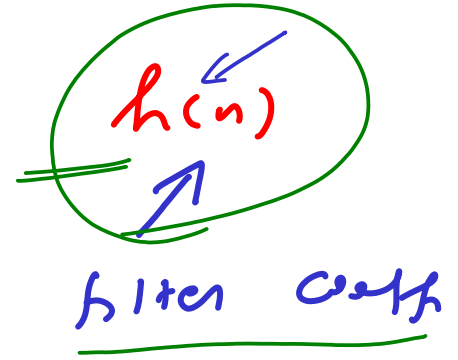
$$\begin{matrix} \omega_1 & \rightarrow & \omega_1 \\ \omega_2 & \rightarrow & \omega_2 \\ \omega_3 & \rightarrow & \omega_3 \end{matrix}$$

multiple analog poles $\rightarrow$ single Digital pole
"ALIASING"

IIR filter ← digital

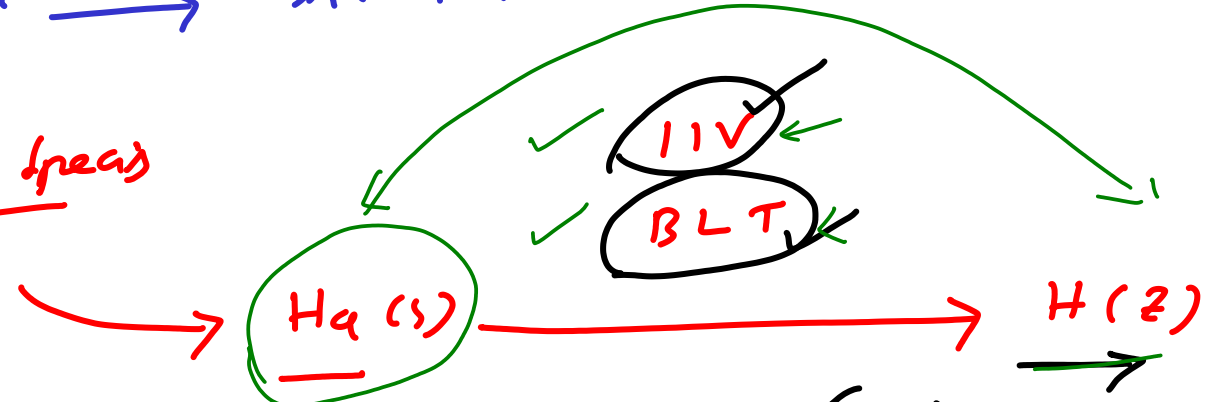
→ Infinite Impulse Response

filter



IIR digital $\xrightarrow{X}$ $h(n) \leftarrow$ infinite duration

Analog spec



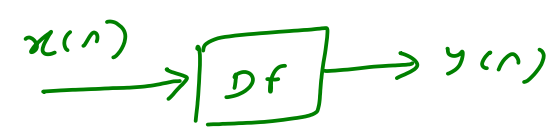
Impulse Invariant Method (IIV)
Bilinear Transformation Tech (BLT)

Structure of IIR

- Direct form - I ✓
- Direct form - II ✓
- Cascaded form - II ✓

LT ← Analog filter ✓
R, L, C

ZT ← Digital filter ✓
Adder, Mult, Scalar



DF - IIR $\Rightarrow$

$$y(n) = - \sum_{k=1}^N a_k \cdot y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

$N \rightarrow$ order of the filter

DIRECT FORM-I

IIR filter $\rightarrow$ recursive filter
 feedback filter

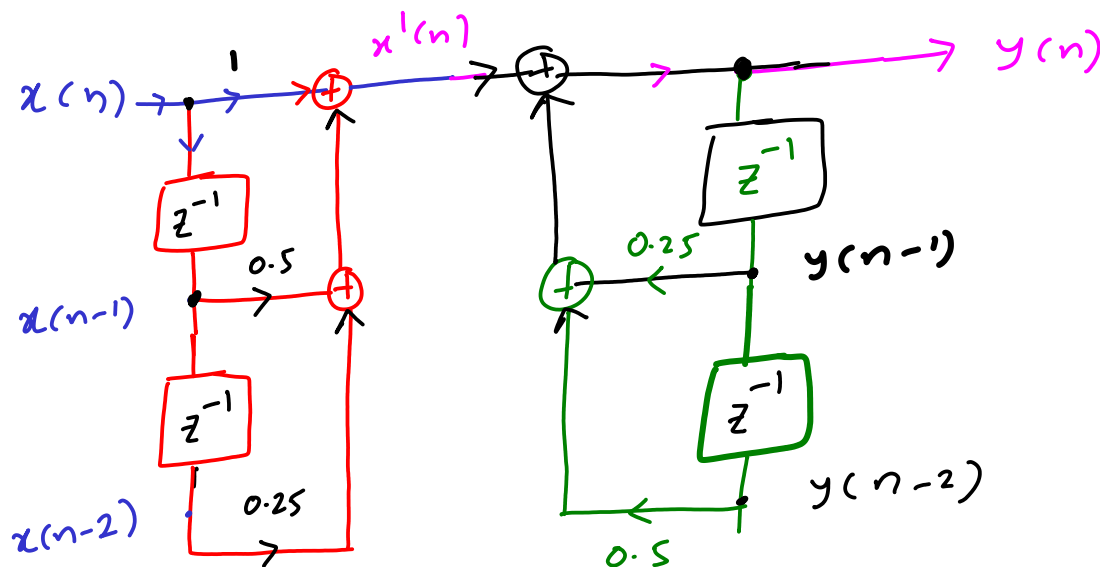
$$y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

$y(n) \rightarrow$ fn [i/p, past o/p]

Time domain Relationship bwn i/p & o/p

- ① obtain Direct form-I Realization of a digital IIR filter described by the following difference equation

$$y(n) = 0.25 y(n-1) + 0.5 y(n-2) + 0.25 x(n-2) + 0.5 x(n-1) + x(n)$$



$z^{-1} \rightarrow$ delay element

④

(2) obtain Direct form I realization of a digital IIR filter with a digital transfer function

$$H(z) = \frac{1 + 0.5z^{-1} + 0.25z^{-2}}{0.25 + 0.5z^{-1} + 0.6z^{-2} + z^{-3}}$$

$$\frac{Y(z)}{X(z)} = \frac{1 + 0.5z^{-1} + 0.25z^{-2}}{0.25 + 0.5z^{-1} + 0.6z^{-2} + z^{-3}}$$

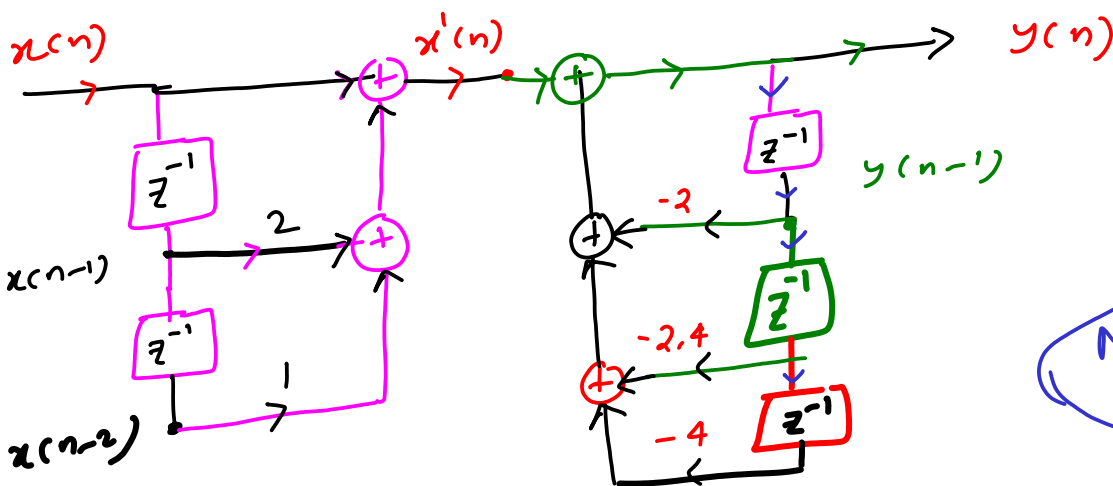
$$Y(z) \cdot 0.25 + 0.5z^{-1}Y(z) + 0.6z^{-2}Y(z) + z^{-3}Y(z) = X(z) + 0.5z^{-1}X(z) + 0.25z^{-2}X(z)$$

(I)

$$z^{-1} [x(n-k)] = z^{-k} X(z)$$

take z^{-1} of eqn (I)

$$y(n) = -2y(n-1) - 2.4y(n-2) - 4y(n-3) + 4x(n) + 2x(n-1) + 1 \cdot x(n-2) \quad \text{--- (II)}$$



5 - delay elements
Memory elements

DIRECT FORM - I

① $I \pm R \rightarrow H(z) = \frac{Y(z)}{X(z)} \rightarrow NR \rightarrow \text{only (ve) powers of 'z'}$
 $\rightarrow dr$

$$y(n) = -2y(n-1) - 2.4y(n-2) - 4y(n-3) + 4x(n) + 2x(n-1) + x(n-2)$$

digital filter,
difference eqn

←
differential eqn $\rightarrow$ Analog filter

DIRECT form - II ?

$$H(z) = \frac{Y(z)}{X(z)}$$

② $H(z) = \frac{Y(z)}{X(z)} = \frac{Y(z)}{W(z)} \cdot \frac{W(z)}{X(z)}$

③ $\frac{W(z)}{X(z)} = \frac{1}{Dr. \text{ polynomial}} \leftarrow \text{structure}$

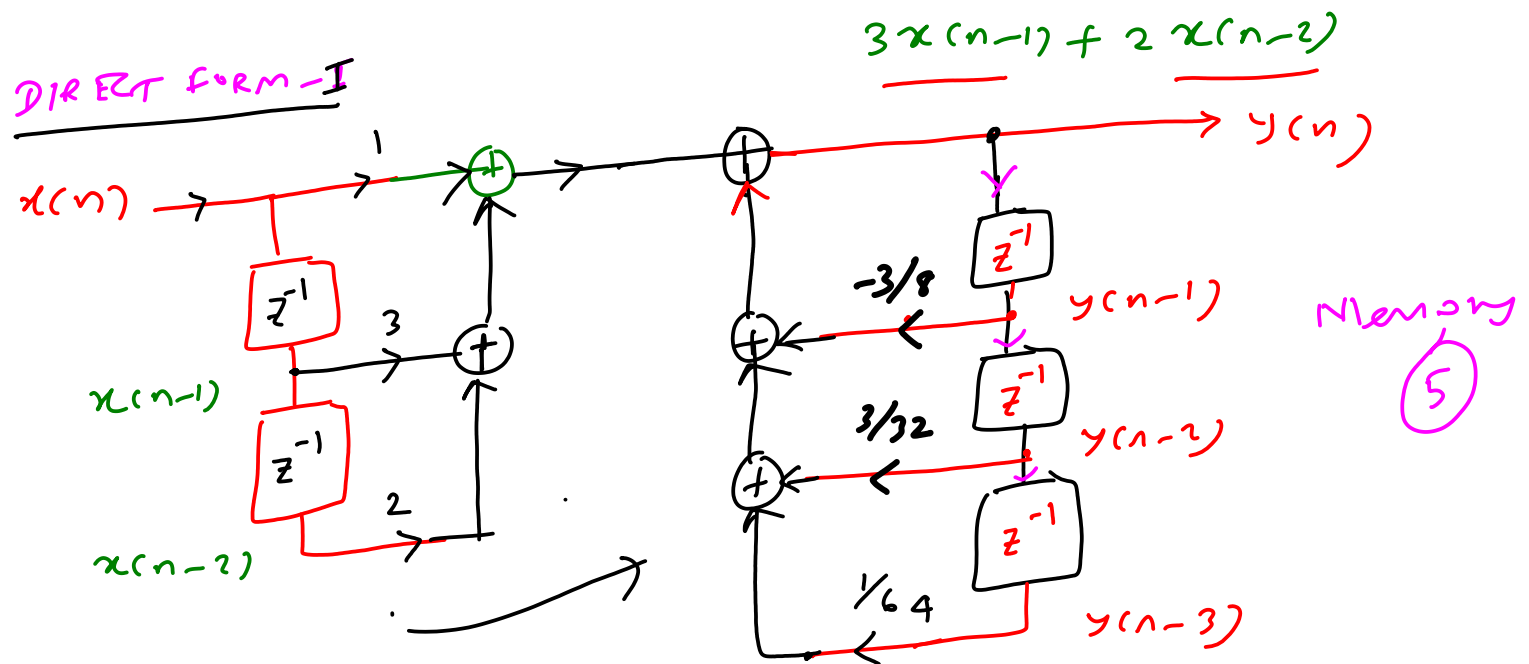
④ $\frac{Y(z)}{W(z)} = Nr. \text{ polynomial} \leftarrow \text{structure}$

⑤ combine str 1 & str 2 $\rightarrow$ overall structure

② Obtain the Direct form I, direct form-II realisations of the LTI system governed by the equation

$$y(n) = -\frac{3}{8} y(n-1) + \frac{3}{32} y(n-2) + \frac{1}{64} y(n-3) + \underline{x(n)} + \underline{3x(n-1)} + \underline{2x(n-2)}$$

DIRECT FORM-I



DIRECT FORM-II

$$y(n) = -\frac{3}{8} y(n-1) + \frac{3}{32} y(n-2) + \frac{1}{64} y(n-3) + \underline{x(n)} + \underline{3x(n-1)} + \underline{2x(n-2)} \quad \text{--- (I)}$$

$$\begin{aligned} x(n) &\xrightarrow{\text{ZT}} X(z) \\ y(n) &\xrightarrow{\text{ZT}} Y(z) \\ x(n-k) &\xrightarrow{\text{ZT}} z^{-k} X(z) \end{aligned}$$

Apply ZT on both sides of eqn (I)

$$Y(z) = -\frac{3}{8} z^{-1} Y(z) + \frac{3}{32} z^{-2} Y(z) + \frac{1}{64} z^{-3} Y(z) + X(z) + 3z^{-1} X(z) + 2z^{-2} X(z)$$

$$Y(z) + \frac{3}{8} z^{-1} Y(z) - \frac{3}{32} z^{-2} Y(z) - \frac{1}{64} z^{-3} Y(z) = X(z) + 3z^{-1} X(z) + 2z^{-2} X(z)$$

$$Y(z) \left[1 + \frac{3}{8} z^{-1} - \frac{3}{32} z^{-2} - \frac{1}{64} z^{-3} \right] = X(z) \left[1 + 3z^{-1} + 2z^{-2} \right]$$

$$\therefore \frac{Y(z)}{X(z)} = H(z) = \frac{1 + 3z^{-1} + 2z^{-2}}{1 + \frac{3}{8} z^{-1} - \frac{3}{32} z^{-2} - \frac{1}{64} z^{-3}} \quad \text{--- (i)}$$

$$\frac{Y(z)}{X(z)} = \frac{Y(z)}{W(z)} \cdot \frac{W(z)}{X(z)} \quad \text{--- (ii)}$$

$$\frac{W(z)}{X(z)} = \frac{1}{1 + \frac{3}{8} z^{-1} - \frac{3}{32} z^{-2} - \frac{1}{64} z^{-3}} \quad \text{--- (iv)}$$

$$\boxed{\frac{Y(z)}{W(z)} = 1 + 3z^{-1} + 2z^{-2}} \quad \text{--- (v)}$$

Consider eqn (4)

$$W(z) + \frac{3}{8} z^{-1} W(z) - \frac{3}{32} z^{-2} W(z) - \frac{1}{64} z^{-3} W(z) = X(z) \quad \text{--- (vi)}$$

Take z^{-1} of eqn (vi)

$$W(n) + \frac{3}{8} W(n-1) - \frac{3}{32} W(n-2) - \frac{1}{64} W(n-3) = X(n)$$

$$\therefore \checkmark W(n) = X(n) - \frac{3}{8} W(n-1) + \frac{3}{32} W(n-2) + \frac{1}{64} W(n-3) \quad \text{--- (vii)}$$

Consider eqn (v)

Take z^{-1} of eqn (v)

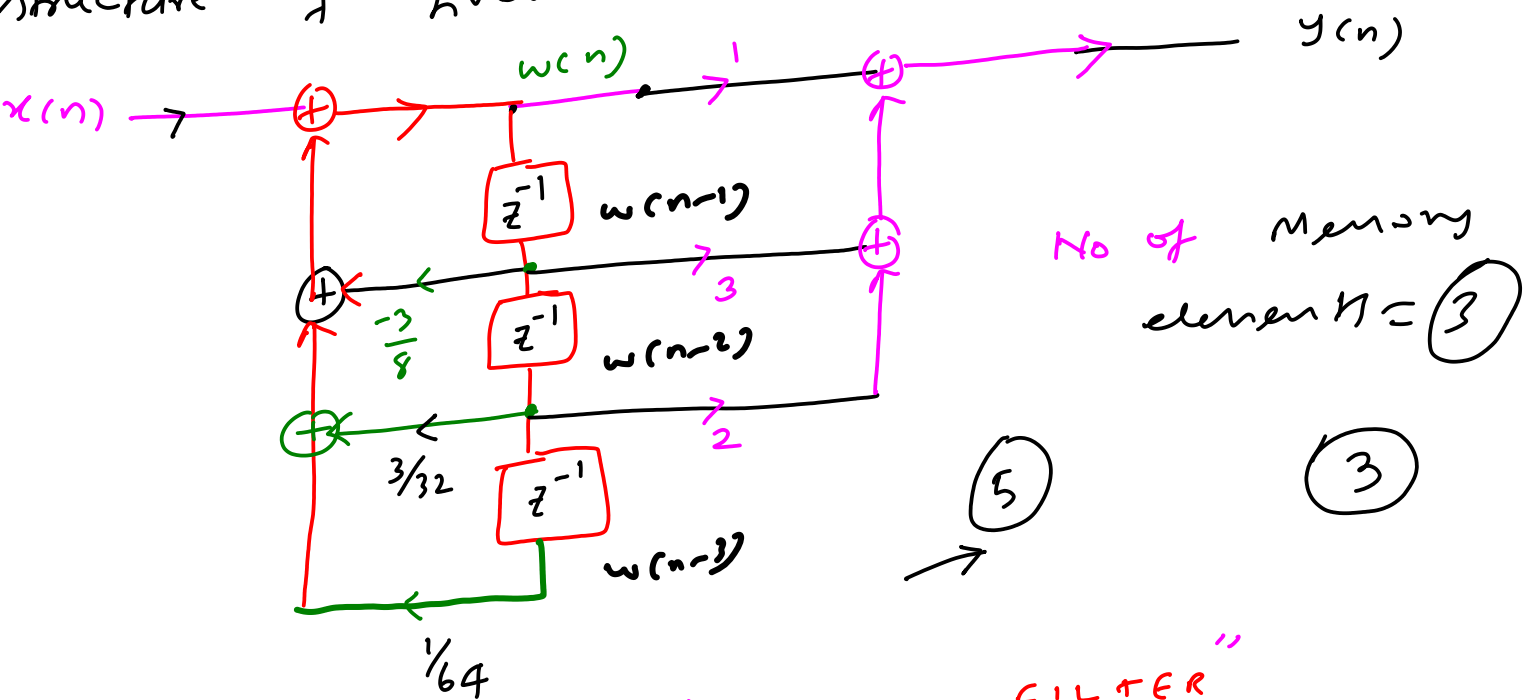
$$\frac{Y(z)}{W(z)} = 1 + 3z^{-1} + 2z^{-2}$$

$$Y(z) = \underline{W(z)} + 3 \underline{z^{-1} W(z)} + 2 \underline{z^{-2} W(z)} \quad \text{--- (viii)}$$

Take z^{-1} of eq (viii)

$$y(n) = w(n) + 3w(n-1) + 2w(n-2) \quad \text{--- (ix)}$$

$\therefore$ Consider eq (7) & (9) to obtain the Direct form II structure of given IIR filter



IIR Realization $\rightarrow$ "RECURSIVE FILTER"

$$y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

$N \rightarrow$ order of the filter $\leftarrow$ No of delay assoc- with o/p

Direct Form - I $\Rightarrow N =$ No of delay elements in the SH

$\rightarrow$ CANONICAL FORM

$N \neq$ No of delay elements present in the structure

→ Non-Canonical form

In the above solved prob

D-II → $N=3$

No of Memory delay = 3 →

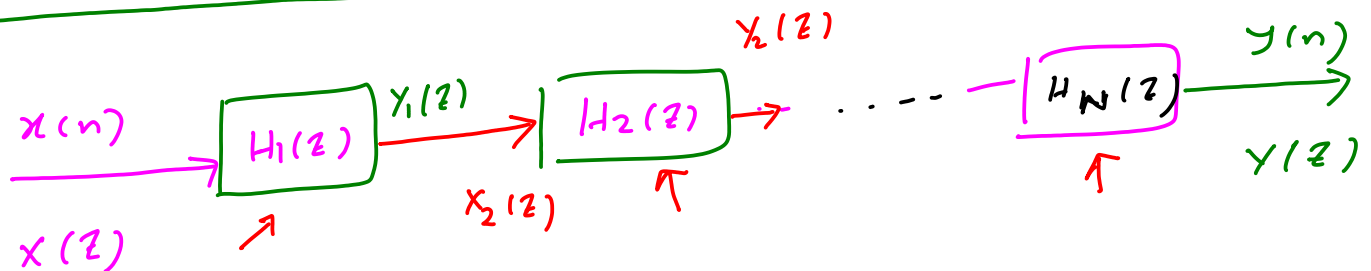
CANONICAL

D-I → $N=3$

No of Memory delay → 5

$N \neq 5$ → Non-Canonical str

III - CASCADE REALIZATION



$H(z) \rightarrow$ overall system function

$$H(z) = H_1(z) \cdot H_2(z) \cdot H_3(z) \cdots H_N(z)$$

IIR - should be described as in terms

$$\frac{Y(z)}{X(z)} = H(z) = \underbrace{H_1(z)}_{\downarrow} \cdot \underbrace{H_2(z)}_{\downarrow} \cdots$$

① Realise the given system in cascaded form.

$$H(z) = \frac{1 + 0.25z^{-1}}{(1 - 2z^{-1} + 0.25z^{-2})(1 - 3z^{-1} + 0.2z^{-2})}$$

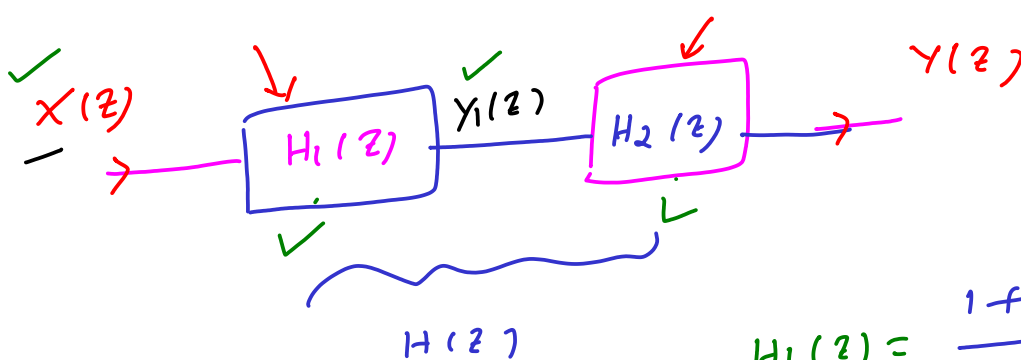
Soln

$$H(z) = \underline{H_1(z)} \cdot \underline{H_2(z)} \quad \text{--- (1)}$$

here

$$H_1(z) = \frac{1 + 0.25z^{-1}}{1 - 2z^{-1} + 0.25z^{-2}}$$

$$H_2(z) = \frac{1}{1 - 3z^{-1} + 0.2z^{-2}}$$



$\oplus$
D-E
✓

$$H_1(z) = \frac{1 + 0.25z^{-1}}{1 - 2z^{-1} + 0.25z^{-2}} \quad \text{--- (1)}$$

Realize of $H_1(z)$ ✓

$$H_1(z) = \frac{y_1(z)}{x(z)} = \frac{y_1(z)}{w_1(z)} \cdot \frac{w_1(z)}{x(z)}$$

$$\frac{w_1(z)}{x(z)} \rightarrow \frac{1}{\text{dr. poly}} \Rightarrow \frac{w_1(z)}{x(z)} = \frac{1}{1 - 2z^{-1} + 0.25z^{-2}} \quad \text{--- (2)}$$

$$\frac{y_1(z)}{w_1(z)} \rightarrow \text{nr. poly} \Rightarrow \frac{y_1(z)}{w_1(z)} = 1 + 0.25z^{-1} \quad \text{--- (3)}$$

Consider eqn (2)

$$w_1(z) - 2z^{-1}w_1(z) + 0.25z^{-2}w_1(z) = x(z)$$

$$w_1(n) - 2w_1(n-1) + 0.25w_1(n-2) = x(n)$$

$$w_1(n) = x(n) + 2w_1(n-1) + 0.25w_1(n-2) \quad \text{--- (4)}$$

Consider eqn (3)

$$Y_1(z) = W_1(z) + 0.25 z^{-1} W_1(z)$$

$$\therefore \checkmark \underline{Y_1(n) = W_1(n) + 0.25 W_1(n-1)} \quad \text{--- (5)}$$

Consider

$$H_2(z) = \frac{1}{1 - 3z^{-1} + 0.2z^{-2}}$$

$$H_2(z) = \frac{Y(z)}{Y_1(z)} = \frac{Y(z)}{W_2(z)} \cdot \frac{W_2(z)}{Y_1(z)}$$

$$\frac{W_2(z)}{Y_1(z)} = \frac{1}{1 - 3z^{-1} + 0.2z^{-2}} \quad \text{--- (6)}$$

$$\frac{Y_1(z)}{W_2(z)} = 1 \quad \text{--- (7)}$$

Consider eqn (1)

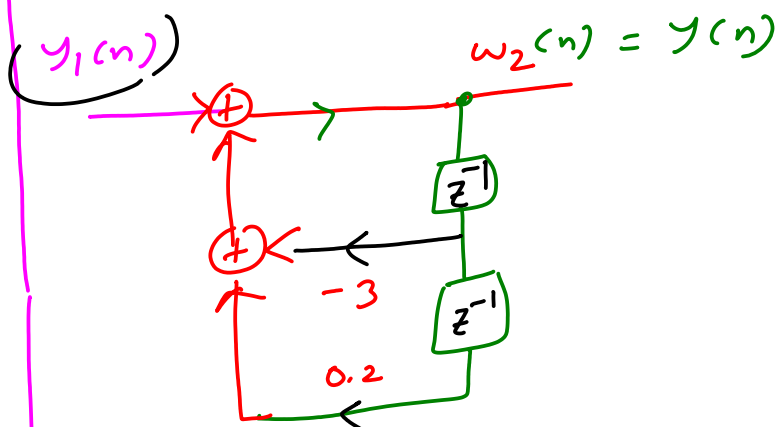
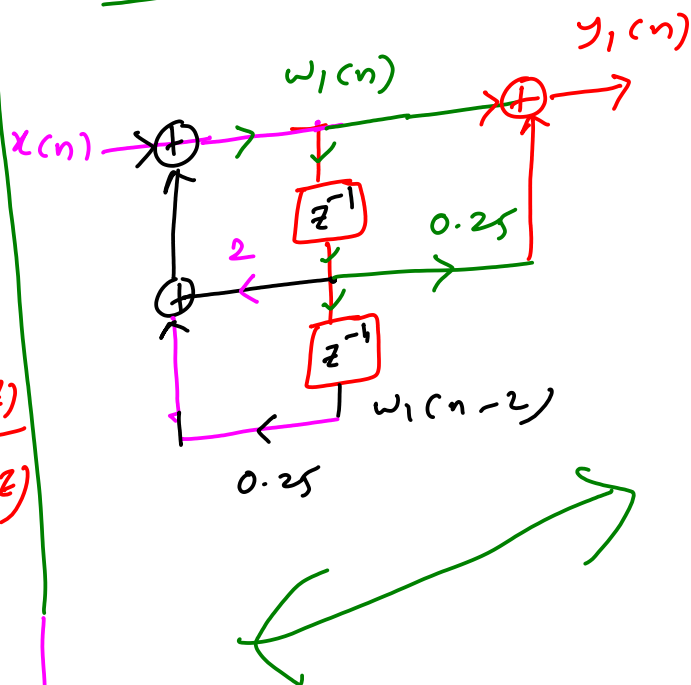
$$W_2(z) - 3z^{-1} W_2(z) + 0.2z^{-2} W_2(z) = Y_1(n)$$

$$\underline{W_2(n) - 3W_2(n-1) + 0.2W_2(n-2) = Y_1(n)}$$

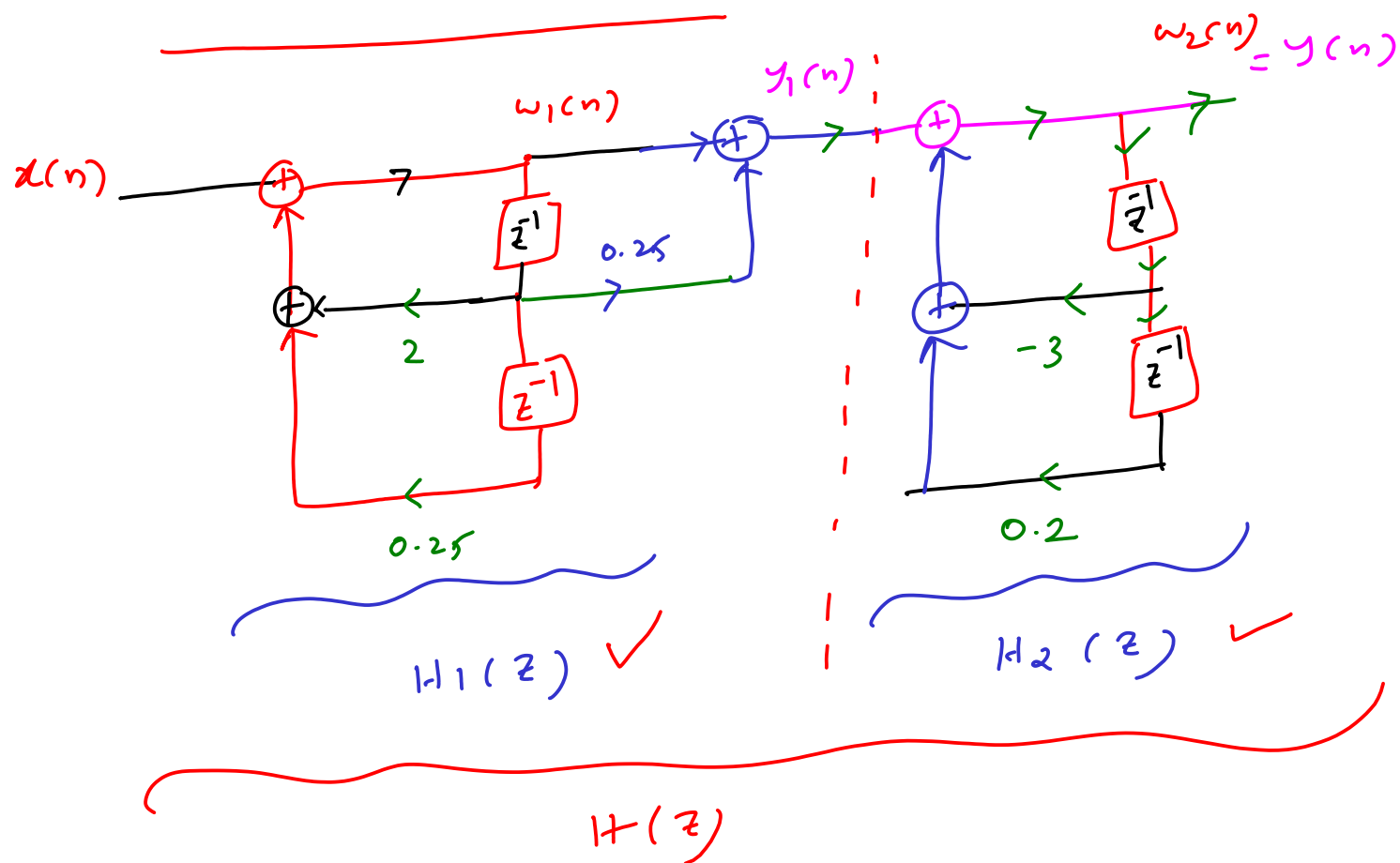
$$\therefore \underline{W_2(n) = Y_1(n) + 3W_2(n-1) + 0.2W_2(n-2)} \quad \text{--- (8)}$$

Consider eqn (2)

$$\underline{Y_2(z) = W_2(z) \Rightarrow Y_2(n) = W_2(n)} \quad \text{--- (9)}$$



CASCADED REALIZATION



② Given

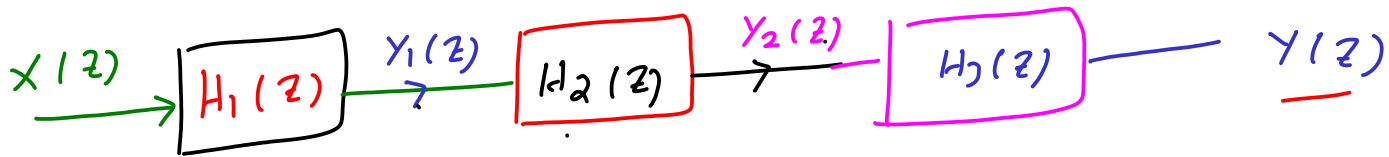
$$H(z) = \frac{2 + 3z^{-1} + 4z^{-2}}{\left(1 + \frac{1}{7}z^{-1}\right) \left(1 - \frac{1}{4}z^{-1}\right) \left(1 + \frac{1}{9}z^{-1}\right)}$$

Obtain cascaded structure.

$$H(z) = H_1(z) \cdot H_2(z) \cdot H_3(z)$$

where $H_1(z) = \frac{2 + 3z^{-1} + 4z^{-2}}{1 + \frac{1}{7}z^{-1}}$

$$H_2(z) = \frac{1}{1 - \frac{1}{4}z^{-1}} \quad ; \quad H_3(z) = \frac{1}{1 + \frac{1}{9}z^{-1}}$$



$$H(z) = H_1(z) \cdot H_2(z) \cdot H_3(z)$$

Consider $H_1(z)$

$$H_1(z) = \frac{2 + 3z^{-1} + 4z^{-2}}{1 + \frac{1}{7}z^{-1}} \quad \leftarrow \text{D-I}$$

$$H_1(z) = \frac{Y_1(z)}{X(z)} = \frac{Y_1(z)}{W_1(z)} \cdot \frac{W_1(z)}{X(z)} ; \text{ here,}$$

$$\frac{W_1(z)}{X(z)} = \frac{1}{\text{dr. poly}} = \frac{1}{1 + \frac{1}{7}z^{-1}} \quad \text{--- (1)}$$

$$\frac{Y_1(z)}{W_1(z)} = \text{Nr. poly} = 2 + 3z^{-1} + 4z^{-2} \quad \text{--- (2)}$$

$$W_1(z) + \frac{1}{7}z^{-1}W_1(z) = X(z)$$

$$\downarrow \quad \checkmark \quad \boxed{W_1(n) = X(n) - \frac{1}{7}W_1(n-1)} \quad \text{--- (3) } \checkmark$$

$$Y_1(z) = 2W_1(z) + 3z^{-1}W_1(z) + 4z^{-2}W_1(z)$$

$$\boxed{Y_1(n) = 2W_1(n) + 3W_1(n-1) + 4W_1(n-2)} \quad \text{--- (4) } \checkmark$$

Consider $H_2(z)$

$$H_2(z) = \frac{1}{1 - \frac{1}{4}z^{-1}} = \frac{Y_2(z)}{Y_1(z)}$$

$$\frac{Y_2(z)}{Y_1(z)} = \frac{Y_2(z)}{W_2(z)} \cdot \frac{W_2(z)}{Y_1(z)}$$

here $\frac{w_2(z)}{y_1(z)} = \frac{1}{\text{dr. poly}} = \frac{1}{1 - \frac{1}{4} z^{-1}}$

$$w_2(z) - \frac{1}{4} z^{-1} w_2(z) = y_1(z)$$

$$w_2(n) - \frac{1}{4} w_2(n-1) = y_1(n)$$

$$w_2(n) = y_1(n) + \frac{1}{4} w_2(n-1) \quad \text{--- (5) ---}$$

||| $\frac{y_2(z)}{w_2(z)} = \text{Nr. poly} = 1$

$\therefore y_2(z) = w_2(z)$

$$y_2(n) = w_2(n) \quad \text{--- (6) ---}$$

Consider $H_3(z)$

$$H_3(z) = \frac{1}{1 + \frac{1}{9} z^{-1}} = \frac{y(z)}{y_2(z)} = \frac{y(z)}{w_3(z)} \cdot \frac{w_3(z)}{y_2(z)} \quad \checkmark$$

$$\frac{w_3(z)}{y_2(z)} = \frac{1}{\text{dr. poly}} = \frac{1}{1 + \frac{1}{9} z^{-1}}$$

$$w_3(z) + \frac{1}{9} z^{-1} w_3(z) = y_2(z)$$

$$w_3(n) = y_2(n) - \frac{1}{9} w_3(n-1) \quad \text{--- (7) ---}$$

||| $\frac{y(z)}{w_3(z)} = \text{Nr. poly} = 1 \Rightarrow y(z) = w_3(z)$

$$y(n) = w_3(n) \quad \text{--- (8) ---}$$

from eqn ③, ④, ⑤, ⑥, ⑦, ⑧

$$w_1(n) = x(n) - \frac{1}{7} w_1(n-1) \quad \checkmark$$

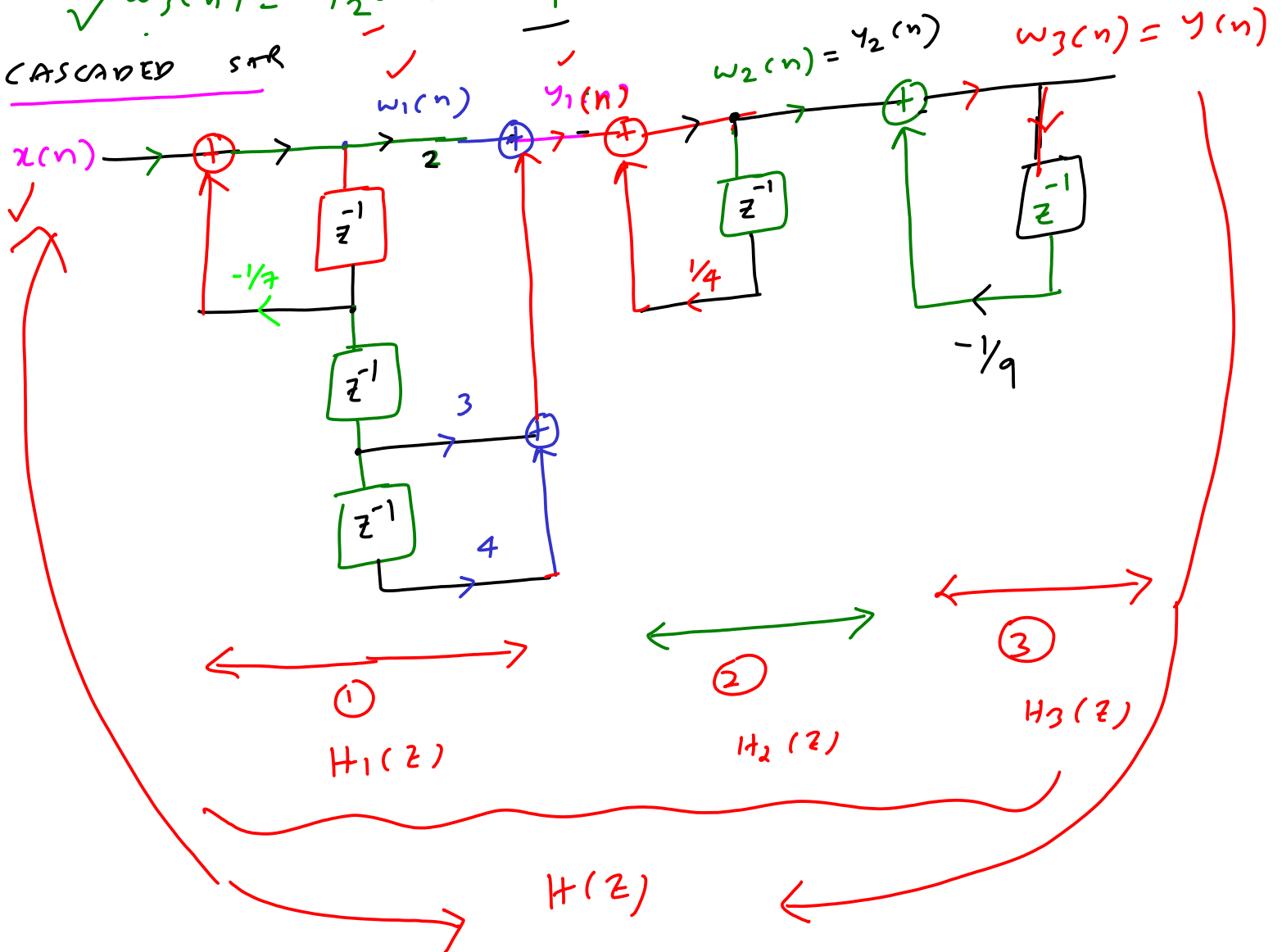
$$y_1(n) = 2 w_1(n) + 3 w_1(n-2) + 4 w_1(n-3) \quad \checkmark$$

$$\checkmark w_2(n) = y_1(n) + \frac{1}{4} w_2(n-1) \quad \checkmark$$

$$y_2(n) = w_2(n) \quad \checkmark$$

$$\checkmark w_3(n) = y_2(n) - \frac{1}{9} w_3(n-1) ; y(n) = w_3(n)$$

CASCADDED SRR



→ diff bwn AF & DF

→ Relation bwn LT & ZT

→ Mapping procedure bwn 'S' & 'Z'

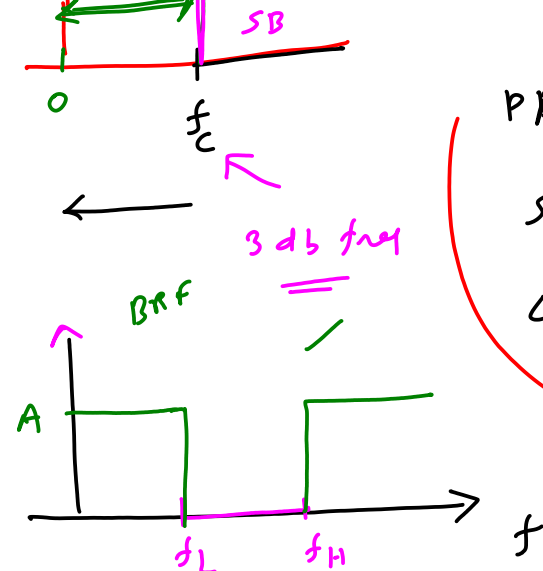
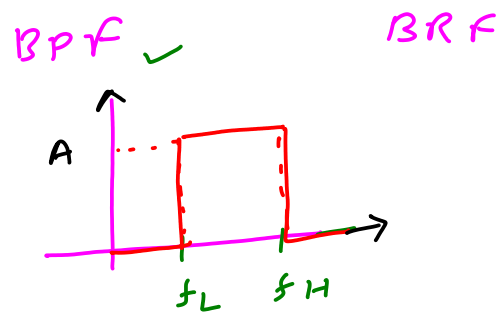
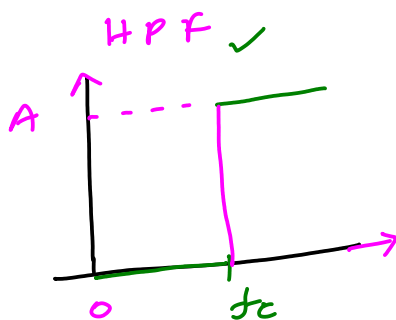
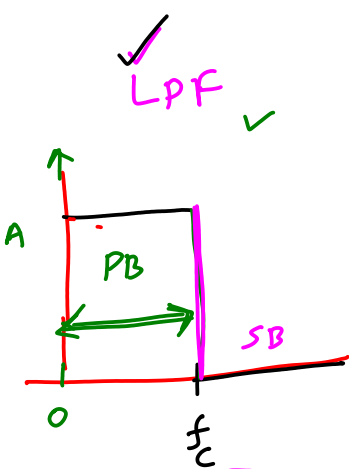
→ structure of IIR $\begin{cases} D-I \\ D-II \\ \text{CASCADABLE} \end{cases}$

→ Design of digital IIR filter $\rightarrow$ $H_a(s)$ $\rightarrow$ $H(z)$ (or) $h(n)$
SPECIFICATIONS

Pass Band $\leftarrow$ PB ✓
Stop Band $\leftarrow$ SB ✓
PB atten $\leftarrow$ α_p ✓
SB atten $\leftarrow$ α_s ✓
PB ripple $\leftarrow$ δ_p ✓
SB ripple $\leftarrow$ δ_s ✓

"PROCEDURE" $\rightarrow$ $H_a(s) \rightarrow H(z)$

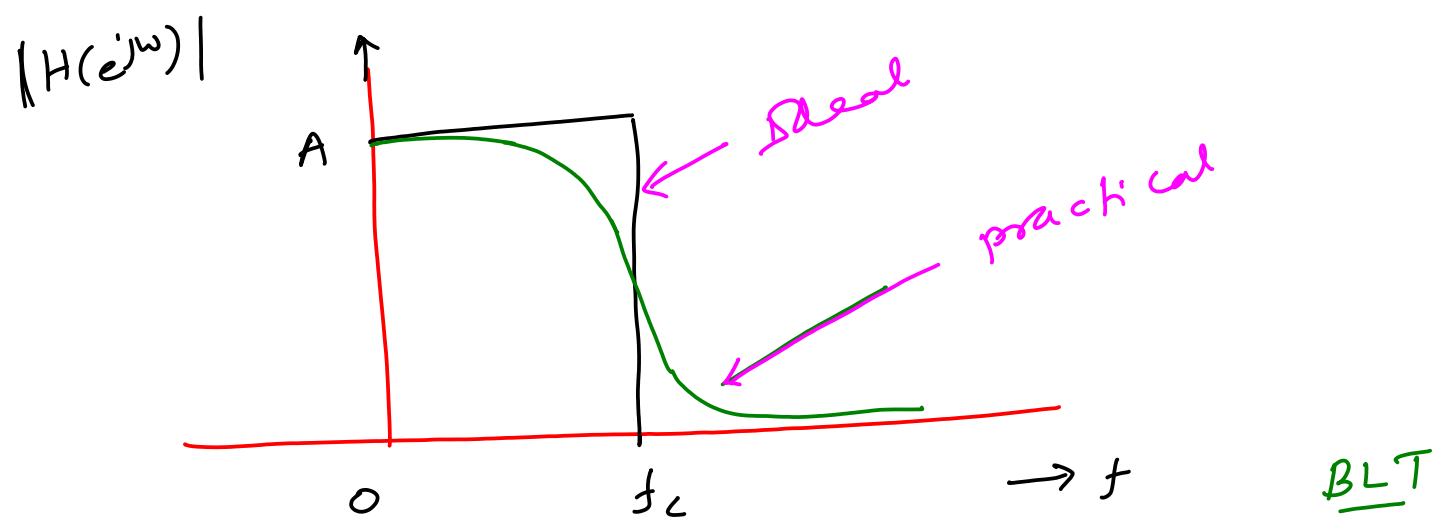
filter $\rightarrow$ Frequency Selective N/w



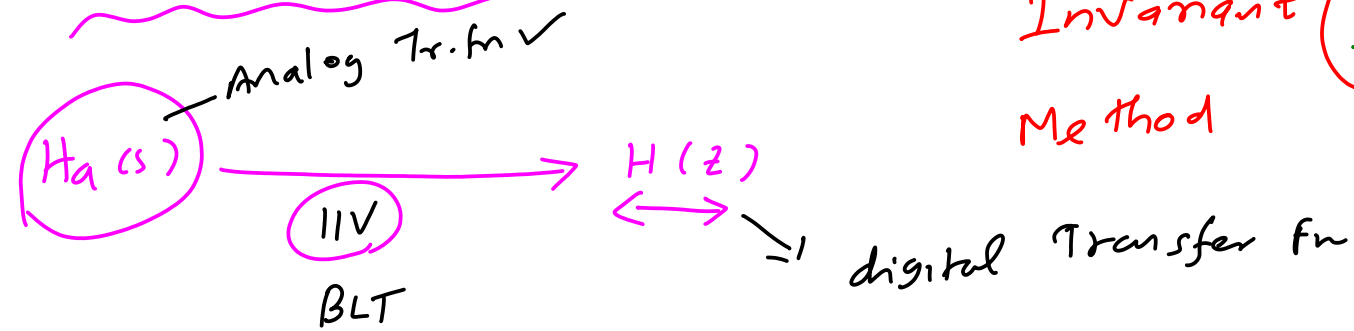
PASS BAND $\rightarrow$ gain $\rightarrow$ const

STOP BAND $\rightarrow$ gain $\rightarrow$ '0'

CUT OFF FREQUENCY $\rightarrow$ freq at which there is a change PB - SB



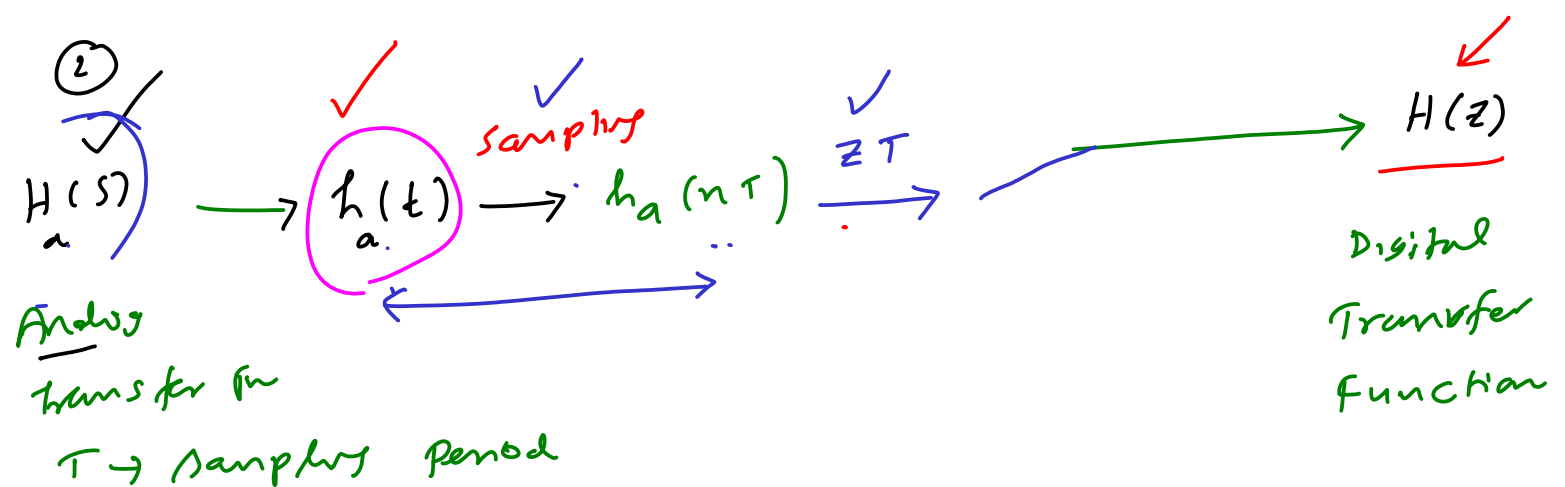
Design of Digital IIR filter - Impulse Invariant (IIV) Method



Impulse Invariant Method (IIV)

① 's' plane $\rightarrow$ 'z' plane

Mapping procedure bwn s-plane & z-plane



$$H_a(s) = \frac{\text{Nr. poly}}{\text{Dr. poly}} = \frac{Y(s)}{X(s)} = \frac{1}{s+p_1} + \frac{1}{s+p_2} + \dots + \frac{1}{s+p_N}$$

$$\frac{1}{s^2+3s+2} = \frac{1}{(s+2)(s+1)} = \frac{k_1}{s+2} + \frac{k_2}{s+1}$$

$N \rightarrow \text{No. of Poles}$

$$H_a(s) = \sum_{i=1}^N \frac{A_i}{s+p_i} = \frac{A_1}{s+p_1} + \frac{A_2}{s+p_2}$$

① $\frac{1}{s+a} \rightarrow e^{-at}$

$$h_a(t) = L^{-1}[H_a(s)] = L^{-1}\left[\sum_{i=1}^N \frac{A_i}{s+p_i}\right]$$

$$= L^{-1}\left[\sum_{i=1}^N \frac{A_i}{s+p_i}\right]$$

$$= \sum_{i=1}^N L^{-1}\left(\frac{A_i}{s+p_i}\right)$$

$$h_a(t) = \sum_{i=1}^N A_i \cdot e^{-p_i t}$$

②

Sampling

$$t = nT$$

$$h_a(nT) = \sum_{i=1}^N A_i \cdot e^{-p_i nT}$$

③

Z-Transform

$$ZT[h_a(nT)] = ?$$

$$ZT[x(n)] = X(z) = \sum_{n=0}^{\infty} x(n) z^{-n} \rightarrow$$

$$= \sum_{n=0}^{\infty} h_a(nT) z^{-n} = \sum_{n=0}^{\infty} \left(\sum_{i=1}^N A_i e^{-p_i nT} \right) z^{-n}$$

$$= \sum_{i=1}^N A_i \sum_{n=0}^{\infty} e^{-p_i nT} z^{-n}$$

$$= \sum_{i=1}^N A_i \sum_{n=0}^{\infty} \left(e^{-p_i T} z^{-1} \right)^n$$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$$

$$H(z) = \sum_{i=1}^N A_i \left[\frac{1}{1 - e^{-p_i T} z^{-1}} \right] \xrightarrow{\text{N}} \text{I}$$

$$H_a(s) = \sum_{i=1}^N A_i \frac{1}{s + p_i}$$

$$\frac{1}{s + p_i} \rightarrow \frac{1}{1 - e^{-p_i T} z^{-1}} ; p_i \rightarrow i^{\text{th}} \text{ pole}$$

$i \rightarrow 1 \text{ to } N$

$$\frac{1}{s - p_i} \rightarrow \frac{1}{1 - e^{p_i T} z^{-1}}$$

$$\frac{s + a}{(s + a)^2 + b^2} \rightarrow \frac{1 - e^{-aT} (\cos bT) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + e^{-2aT} z^{-2}}$$

write down the procedure involved in IV method to design digital IIR filter.

$$\frac{b}{(s + a)^2 + b^2} \rightarrow \frac{e^{-aT} (\sin bT) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + e^{-2aT} z^{-2}}$$

while Sampling $t = nT$

$T \rightarrow$ sampling period

$$T < 1 \mu c \rightarrow$$

$$T \geq 1 \mu c \rightarrow$$

$$H_N(z) = T \cdot H(z)$$

$$H_N(z) = H(z)$$

① for the analog Transfer function

$$H(s) = \frac{2}{s^2 + 3s + 2} \quad \text{design its equivalent}$$

digital filter if a) $T=1 \text{ sec}$, b) $T=0.1 \text{ sec}$

Soln

Given: $H(s) = \frac{2}{s^2 + 3s + 2}$ — (1)

$$\frac{2}{s^2 + 3s + 2} = \frac{2}{(s+2)(s+1)} = \frac{A}{s+2} + \frac{B}{s+1} \quad \text{--- (2)}$$

$$A = \cancel{(s+2)} \left[\frac{2}{\cancel{(s+2)}(s+1)} \right] \text{ at } s = -2 = \textcircled{-2}$$

$$B = \cancel{(s+1)} \left[\frac{2}{(s+2)\cancel{(s+1)}} \right] \text{ at } s = -1 = \textcircled{2}$$

$$H(s) = \frac{2}{(s+2)(s+1)} = \frac{-2}{s+2} + \frac{2}{s+1}$$

$$H(s) = 2 \left[\frac{1}{s+1} - \frac{1}{s+2} \right] \quad \text{--- (3)}$$

$$\frac{1}{s+p_i} \rightarrow \frac{1}{1 - e^{-p_i T} \cdot z^{-1}}$$

$$\frac{1}{s+1} \rightarrow \frac{1}{1 - e^{-1 \cdot T} \cdot z^{-1}}$$

$$\frac{1}{s+2} \rightarrow \frac{1}{1 - e^{-2T} \cdot z^{-1}}$$

$$H(z) = 2 \left[\frac{1}{1 - e^{-T} z^{-1}} - \frac{1}{1 - e^{-2T} z^{-1}} \right] \quad \text{--- (3)}$$

$T \rightarrow$ sampling period

Ex i) $T = 1 \text{ sec}$

$$H(z) = 2 \left[\frac{1}{1 - e^{-1} z^{-1}} - \frac{1}{1 - e^{-2} z^{-1}} \right]$$

$$= 2 \left[\frac{1}{1 - 0.3679 z^{-1}} - \frac{1}{1 - 0.1353 z^{-1}} \right]$$

$$H(z) = \frac{0.4652 z^{-1}}{1 - 0.5032 z^{-1} + 0.0498 z^{-2}}$$

Car i) $T = 0.1 \text{ sec}$

$T < 1 \text{ sec}$

$$H_N(z) = T \cdot H(z) = \text{---} \quad (1)$$

$$H(z) = 2 \left[\frac{1}{1 - e^{-0.1} z^{-1}} - \frac{1}{1 - e^{-0.2} z^{-1}} \right]$$

$$= \frac{2}{1 - 0.9048 z^{-1}} - \frac{2}{1 - 0.8187 z^{-1}}$$

$$= \frac{2(1 - 0.8187 z^{-1}) - 2(1 - 0.9048 z^{-1})}{(1 - 0.9048 z^{-1})(1 - 0.8187 z^{-1})}$$

$$H(z) = \frac{0.1722 z^{-1}}{1 - 1.7235 z^{-1} + 0.7408 z^{-2}} ; \quad \text{but here}$$

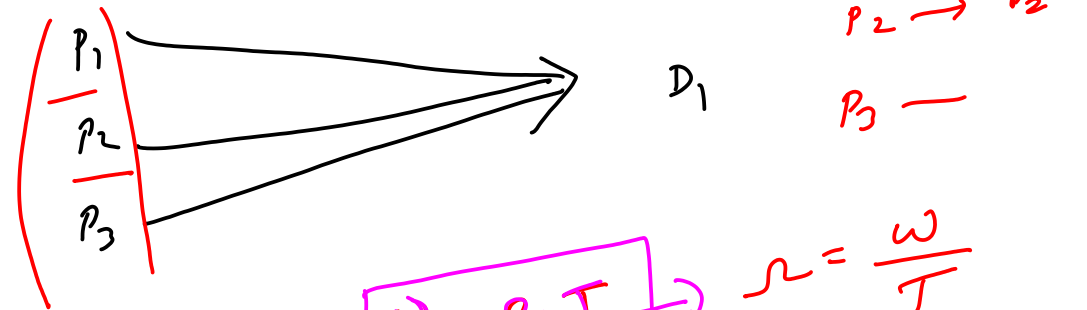
$$T = 0.1 \text{ sec} < 1$$

$$H_N(z) = T \cdot H(z) = \frac{0.01722 z^{-1}}{1 - 1.7235 z^{-1} + 0.7408 z^{-2}}$$

Mapping bwn s - z plane $\rightarrow$ aliasing

Non-Confidential
Mapping
Many to one

Many analog poles can be mapped into a single digital pole



$$\boxed{\omega = \Omega T} \Rightarrow \Omega = \frac{\omega}{T}$$

$T \rightarrow$ sampling period

② Convert analog filter with a system Transfer function

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9} \quad \text{into a}$$

digital IIR filter by means of IIV method

soln

$$\frac{s + a}{(s + a)^2 + b^2} \rightarrow \frac{1 - e^{-aT} (\cos bT) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + e^{-2aT} z^{-2}}$$

$$\frac{s+0.1}{(s+0.1)^2 + 3^2} \rightarrow \frac{1 - e^{-0.1T} \cos(3T) z^{-1}}{1 - 2e^{-0.1T} \cos(3T) z^{-1} + e^{-0.2T} z^{-2}}$$

$a = 0.1$
 $b = 3$

here T is NOT Given

Assume $T = 1 \text{ sec}$

$$H(z) = \frac{1 - e^{-0.1} \cos(3) z^{-1}}{1 - 2e^{-0.1} \cos(3) z^{-1} + e^{-0.2} z^{-2}}$$

$$H(z) = \frac{1 + 0.8958 z^{-1}}{1 + 1.7916 z^{-1} + 0.8187 z^{-2}}$$

Carx $T = 0.1 \text{ sec} \checkmark$

$$H(z) = \frac{1 - e^{-0.01} \cos(0.3) z^{-1}}{1 - 2e^{-0.01} \cos(0.3) z^{-1} + e^{-0.02} z^{-2}}$$

$$H(z) =$$

$$H_H(z) = 0.1 (H(z)) =$$

?

IIV $\rightarrow$ s - z Mapping

$$\rightarrow h_a(s) \rightarrow h_a(t) \rightarrow h_a(nT) \rightarrow H(z)$$

$$\rightarrow \frac{1}{s + p_i} \rightarrow$$

$$\rightarrow \frac{1}{s - p_i} \rightarrow$$

$$\rightarrow \frac{s + a}{(s + a)^2 + b^2} \rightarrow$$

$$\frac{b}{(s + a)^2 + b^2} \rightarrow$$

Non-conformal
mapping

$$\rightarrow H_N(z) \quad ?$$

$$\rightarrow \omega = \Omega T$$

$\rightarrow$ drawback of IIV-method

BILINEAR TRANSFORMATION (BLT)

"CONFORMAL MAPPING"



ONE-ONE

$$A_1 \rightarrow D_1$$

$$A_2 \rightarrow D_2$$

"No aliasing"

In BLT

$$H(z) = H(s) \text{ at}$$

$$s = \frac{2}{T} \left[\frac{1 - z^{-1}}{1 + z^{-1}} \right]$$

$$s = \sigma + j\omega$$

Relation bwn 's' & 'ω'

put $s = j\omega$, $z = e^{j\omega T}$

$$j\omega = \frac{2}{T} \left[\frac{1 - e^{-j\omega T}}{1 + e^{-j\omega T}} \right]$$

$$= \frac{2}{T} \left[\frac{e^{j\omega T/2} e^{-j\omega T/2} - e^{-j\omega T}}{e^{j\omega T/2} e^{-j\omega T/2} - e^{-j\omega T}} \right]$$

$$= \frac{2}{T} \frac{e^{-j\omega T/2} \left[e^{j\omega T/2} - e^{-j\omega T/2} \right]}{e^{-j\omega T/2} \left[e^{j\omega T/2} - e^{-j\omega T/2} \right]}$$

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$j\omega = \frac{2}{T} \frac{2j \sin \omega T/2}{2 \cos \omega T/2}$$

$$\Omega = \frac{2}{T} \frac{\sin \omega/2}{\cos \omega/2}$$

BLT

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$$

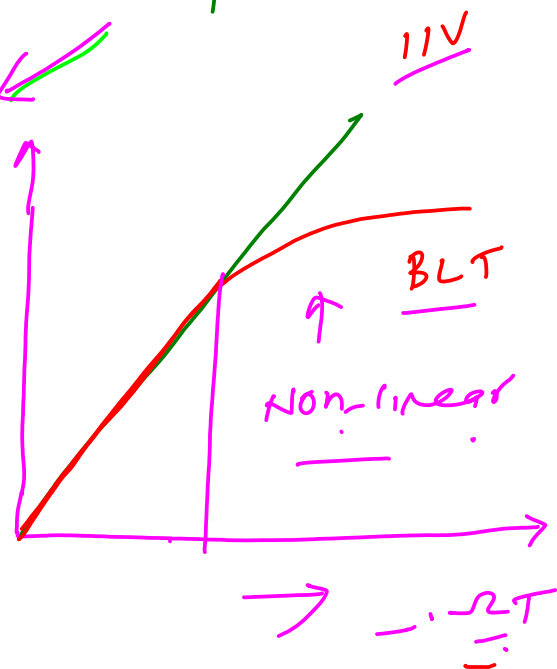
11V ✓

$$\omega = \Omega T$$

$$\Omega = \frac{\omega}{T}$$

$$\omega = 2 \cdot \tan^{-1}\left(\frac{\Omega T}{2}\right)$$

$$y = mx + c$$



Warping effect

=

amplitude distortion

What is

meant by warping effect

$\Omega_p \rightarrow$ Pre-warping freq

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right)$$

Warping effect

Can be eliminated

11V ✓

BLT ✓

① Many to ONE
(Non-Conformal)

① 1-1 Mapping (CONFIRMAL)

② Aliasing

② No Aliasing

③ $\omega = \Omega \cdot T$

③ $\omega = 2 \tan^{-1} \left(\frac{\Omega T}{2} \right)$

④ $\sum_{i=1}^N \frac{A_i}{s - p_i} = \sum_{i=1}^N \frac{A_i}{1 - e^{\frac{p_i T}{2}} z^{-1}}$

④ $s = \frac{2}{T} \left[\frac{1 - z^{-1}}{1 + z^{-1}} \right]$

Pbm 3 for the analog transfer function

$H(s) = \frac{2}{s^2 + 3s + 2}$, determine $H(z)$

Using BLT. Assume that $T = 1 \mu\text{s}$, $T = 0.1 \text{ sec}$

Soln In BLT

$H(z) = H(s)$ at $s = \frac{2}{T} \left[\frac{1 - z^{-1}}{1 + z^{-1}} \right]$

$H(s) = \frac{2}{s^2 + 3s + 2}$

$$\tau = 1 \mu s$$

$$H(z) = \frac{2}{\left[\left\{ \frac{2}{\tau} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] \right\}^2 + 3 \left(\frac{2}{\tau} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 2 \right) \right]}$$

$$= \frac{2}{4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 6 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 2}$$

$$= \frac{2 (1+z^{-1})^2}{4 (1-z^{-1})^2 + 6 (1-z^{-1})(1+z^{-1}) + 2 (1+z^{-1})^2}$$

$$H(z) = \frac{0.1667 + 0.3333z^{-1} + 0.1667z^{-2}}{1 - 0.3333z^{-1}}$$

$$\tau = 0.1 \mu s$$

$$H(z) = \frac{0.6043 + 0.0087z^{-1} + 0.0043z^{-2}}{1 - 1.7229z^{-1} + 0.7403z^{-2}}$$

Impulse Invariant Method

$$\omega = \Omega T$$

$T \rightarrow$ Sampling period

$$\frac{1}{\text{step}} \rightarrow$$

BILINEAR TRANSFORMATION TECHNIQUE

$$s = \frac{2}{T} \left[\frac{1 - z^{-1}}{1 + z^{-1}} \right]$$

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right); \quad \begin{matrix} \omega \rightarrow \text{DF} \rightarrow \text{rad/sec} \\ \Omega \rightarrow \text{AF} \end{matrix}$$

$H(s) \rightarrow$ Given directly

Design - Type - I

$H(s) \rightarrow$ Analog Transfer function will not be given directly

Design - II

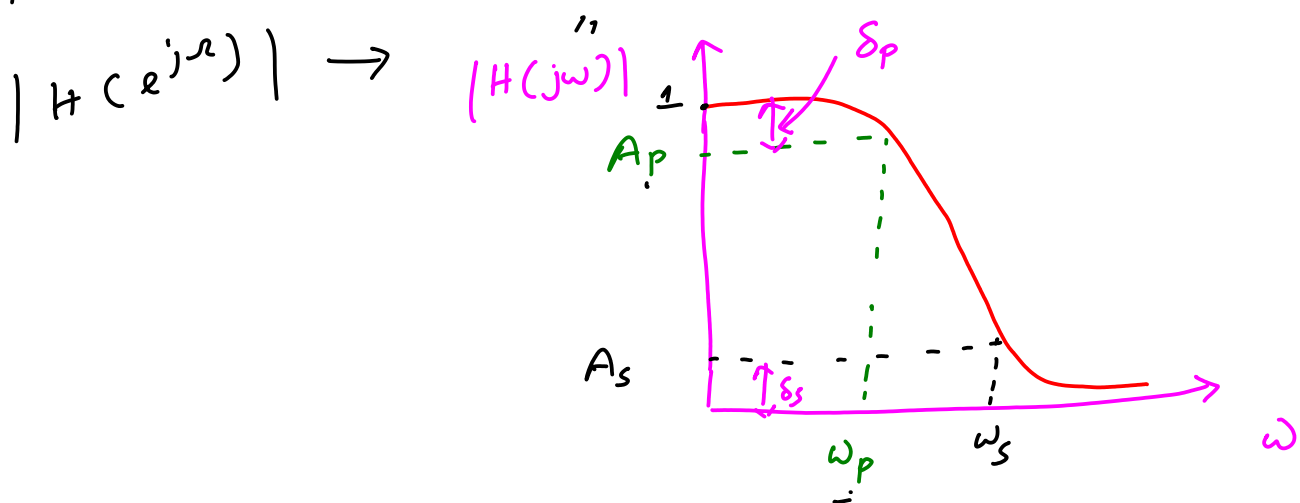
Specifications of IIR Filter

$H(e^{j\omega}) \rightarrow$ Freq. Response of Digital IIR filter

$H(e^{j\Omega}) \rightarrow$ " " " Analog " "

$|H(e^{j\omega})| \rightarrow$ Magnitude Response of Dig. IIR filter

" " " Analog " "



$\omega_p \rightarrow$ PB frequency in rad/sample

✓ $A_p \rightarrow$ PB gain at ' ω_p '

$\omega_s \rightarrow$ SB freq in rad/sample

✓ $A_s \rightarrow$ SB gain at ' ω_s '

✓ $\delta_p \rightarrow$ PB Ripple

✓ $\delta_s \rightarrow$ SB Ripple

✓ $\alpha_p \rightarrow$ PB attenuation

✓ $\alpha_s \rightarrow$ SB attenuation

Normal value
 $A_p \rightarrow$ 'dB' $\rightarrow$ decibel

$$A_{p,db} = 20 \log_{10}(A_p)$$

$$A_p = 10^{A_{p,db}/20}$$

$$A_{s,db} = 20 \log_{10}(A_s)$$

$$A_s = 10^{A_{s,db}/20}$$

$$\alpha_p = \frac{1}{A_p} \Rightarrow \alpha_{p,db} = 20 \log_{10}(\alpha_p) = 20 \log_{10}\left(\frac{1}{A_p}\right)$$
$$= 20 \log_{10}(1) - 20 \log_{10}(A_p)$$

$$\alpha_s = \frac{1}{A_s}$$

$$\alpha_{p,db} = -A_{p,db}$$

Similarly

$$\alpha_{s,db} = -A_{s,db}$$

$$\alpha_{p,db} = \delta_{p,db} = -A_{p,db}$$

$$\alpha_{s,db} = \delta_{s,db} = -A_{s,db}$$

$$A_{p,db} = -\delta_{p,db}$$

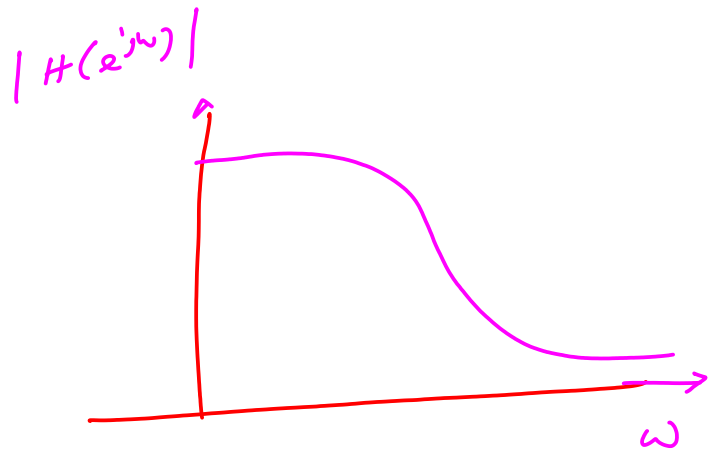
$$20 \log_{10}(A_p) = -\delta_{p,db}$$

$$\boxed{A_p = 10 \quad -\delta_p, \text{db}/20} \rightarrow A_p = 10 \quad -\delta_p, \text{db}/20$$

$$\text{high} \quad \boxed{A_s = 10 \quad -\delta_s, \text{db}/20} \rightarrow A_s = 10 \quad -\delta_s, \text{db}/20$$

$$H(s) = ?$$

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2N}} \quad \text{--- (1)}$$



→ approximated Magnitude Response of
Analogy LPF

$$H(j\omega) H(-j\omega) = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2N}} \quad s = j\omega$$

$$H(s) H(-s) = \frac{1}{1 + \left(\frac{s}{j\omega_c}\right)^{2N}} = \frac{1}{1 + \left[\left(\frac{s}{j\omega_c}\right)^2\right]^N}$$

$$= \frac{1}{1 + \left(-\frac{s^2}{\omega_c^2}\right)^N}$$

$$H(s_n) H(-s_n) = \frac{1}{1 + \left(-s_n^2\right)^N} = \frac{1}{1 + \left(-s_n^2\right)^N}$$

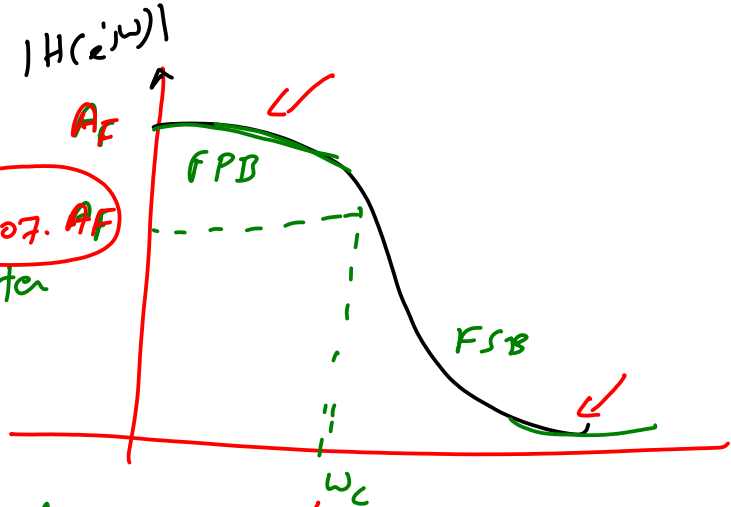
Butterworth LPF

① All poles design

② Flat PB & Flat SB filter

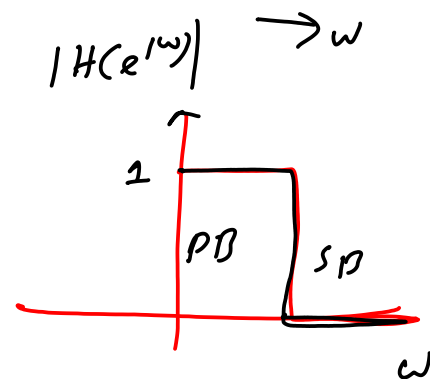
③ Magnitude response of

BW filter is Monotonically
decreasing w.r.t 'w'



④ As $N \rightarrow \infty \rightarrow$ ideal response

⑤ $N \rightarrow$ completely specify the filter



⑥ at $w_c \rightarrow |H(e^{jw})| \rightarrow 70.7\%$ Max. gain

wk off
fr

-3db

← db

0.707

$20 \log_{10}(0.707)$

$2N \rightarrow \text{poles}$

③ $N \rightarrow$ ORDER OF THE FILTER

(1, 3, 5, 7, ...) ODD

$$S_n = e^{j\frac{\pi k}{N}}$$

$k = 1 \text{ to } 2N$

even (2, 4, 6, ...)

$$S_n = e^{\frac{j(2k-1)\pi}{2N}}$$

$k = 1 \text{ to } 2N$

① det the transfer function of Butterworth LPF
if $N=1$

Given

$N=1 \rightarrow \text{odd}$

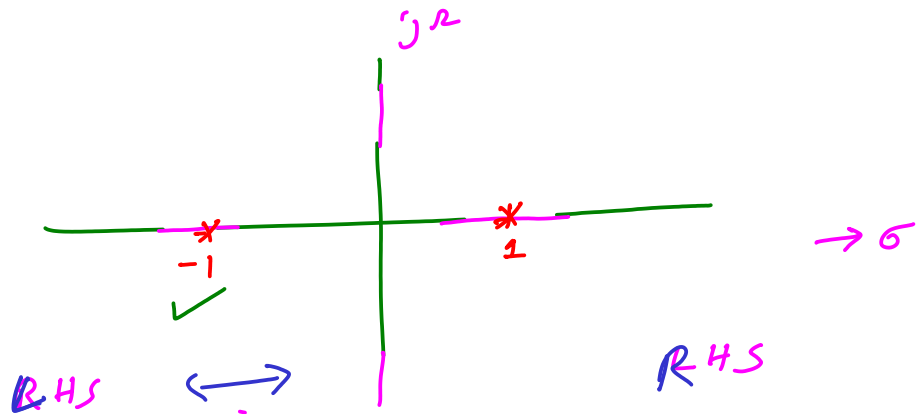
$$s_n = e^{\frac{j\pi k}{N}} ; \quad \begin{matrix} k=1 \text{ to } 2N \\ \underline{k=1,2} \end{matrix}$$

$$\underline{k=1} \rightarrow s_n = e^{\frac{j\pi 1}{1}} = e^{j\pi} = \cos \pi + j \sin \pi = \boxed{-1} \quad \checkmark \quad \frac{1}{s - (-1)}$$

$$k=2 \rightarrow s_n = e^{\frac{j\pi 2}{1}} = e^{j2\pi} = \cos 2\pi + j \sin 2\pi = \boxed{1} \quad \times$$

$$H(s) = \frac{1}{[s_n - (-1)]}$$

$$\boxed{H(s) = \frac{1}{s_n + 1}}$$



② determine transfer function of the Butterworth filter for $N=2$ ✓

Given

$N=2 \rightarrow \text{even}$

$$s_n = e^{\frac{j(2k-1)\pi}{2N}} ; \quad k=1 \text{ to } 2N$$

$$\begin{aligned} k=1 \rightarrow s_n &= e^{\frac{j(2-1)\pi}{4}} = e^{j\pi/4} \\ &= \cos \pi/4 + j \sin \pi/4 \end{aligned} \quad k=1,2,3,4$$

$$s_n = 0.707 + j0.707 \quad \times$$

k=2

$$s_n = e^{j(4-1)\pi/4} = e^{j3\pi/4} = \cos \frac{3\pi}{4} + j \sin \frac{3\pi}{4}$$

$$s_n = -0.707 + j0.707$$

k=3

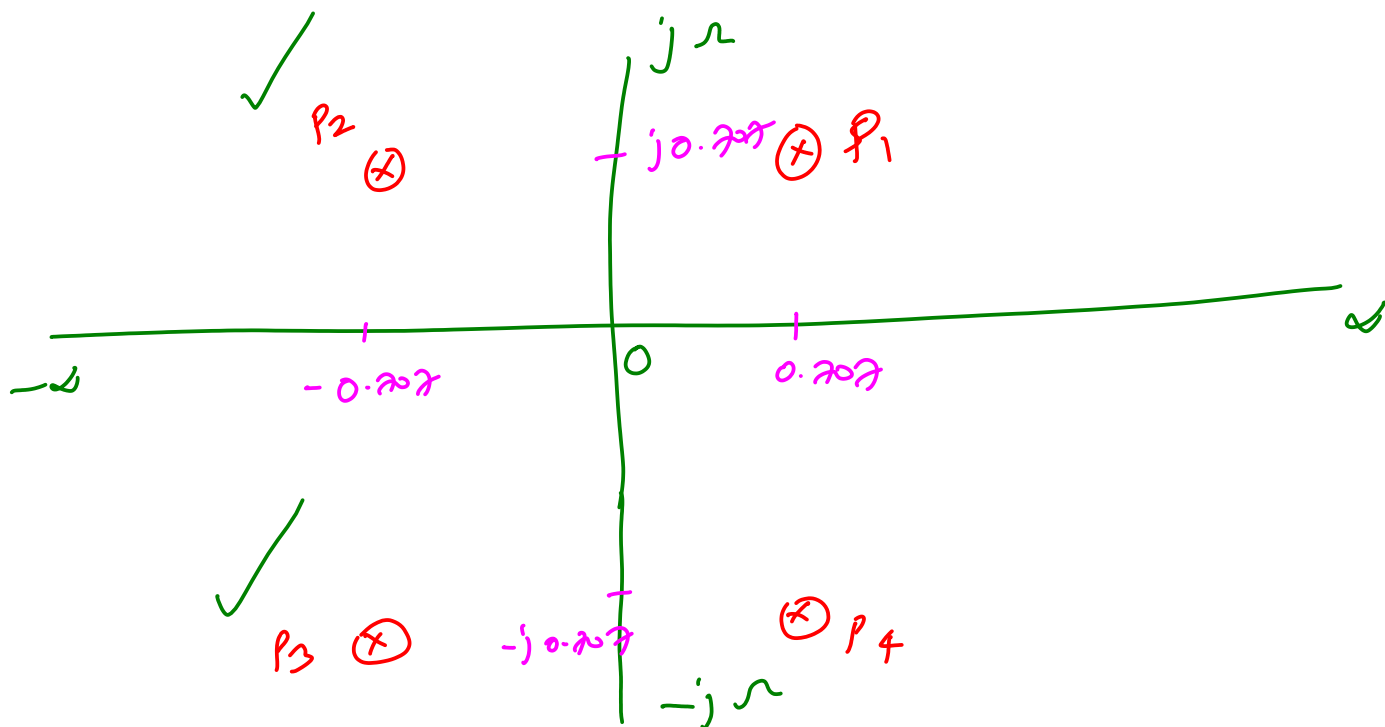
$$s_n = e^{j(6-1)\pi/4} = e^{j5\pi/4} = \cos \frac{5\pi}{4} + j \sin \frac{5\pi}{4}$$

$$s_n = -0.707 - j0.707$$

k=4

$$s_n = e^{j(8-1)\pi/4} = e^{j7\pi/4} = \cos \frac{7\pi}{4} + j \sin \frac{7\pi}{4}$$

$$s_n = 0.707 - j0.707$$



$$\therefore H(s_n) = \frac{1}{(s_n - p_2)(s_n - p_3)}$$

here $p_2 = -0.707 + j0.707$; $p_3 = -0.707 - j0.707$

$$= \frac{1}{\left[(s_n + 0.707) - j0.707 \right] \left[(s_n + 0.707) + j0.707 \right]}$$

$\uparrow$
 a

$$= \frac{1}{(s + 0.707)^2 + (0.707)^2}$$

$$= \frac{1}{s_n^2 + (0.707)^2 + 2 \times 0.707 s_n + (0.707)^2}$$

$$H(s_n) = \frac{1}{s_n^2 + 1.414 s_n + 1}$$

N

1

$$\frac{H(s_n)}{s_n + 1} \quad \checkmark$$

2

$$\frac{1}{s_n^2 + 1.414 s_n + 1} \quad \checkmark$$

3

$$\frac{1}{(s_n + 1)(s_n^2 + s_n + 1)} \quad \checkmark$$

4

$$\frac{1}{(s_n^2 + 0.765 s_n + 1)(s_n^2 + 1.848 s_n + 1)} \quad \checkmark$$

5

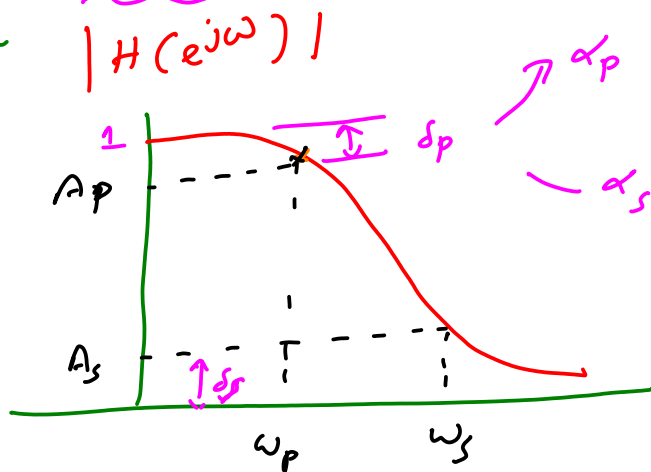
$$\frac{1}{(s_n + 1)(s_n^2 + 0.618 s_n + 1)(s_n^2 + 1.618 s_n + 1)} \quad \checkmark$$

10

Steps involved in Design of Digital IIR filter when its specifications are given

① $N \rightarrow$ specify the filter

② determine the order of the filter (N)



(*)

$$N \geq \frac{1}{2} \cdot \text{Log} \left[\frac{\left(\frac{1}{A_s^2} \right) - 1}{\left(\frac{1}{A_p^2} \right) - 1} \right] \checkmark$$

$A_s \rightarrow$ SB gain

$A_p \rightarrow$ PB gain

$f_s \rightarrow$ Analog SB frequency

$f_p \rightarrow$ Analog PB frequency

$$\text{Log} \left[\frac{f_s}{f_p} \right]$$

If $\alpha_s, (db)$ & $\alpha_p, (db)$ are given, then

$$N \geq \frac{1}{2} \cdot \text{Log} \left[\frac{10^{0.1 \alpha_s, db} - 1}{10^{0.1 \alpha_p, db} - 1} \right]$$

$$N \geq 0.6$$

(*)

$$\text{Log} \left[\frac{f_s}{f_p} \right]$$

Note.

After evaluating the above expression the obtained value $\times$ its next highest integer

(*)

ROUND

$$N \geq 0.01 \rightarrow \boxed{N=1}$$

$$N \geq 1.01 \Rightarrow \boxed{N=2}$$

$$N \geq 0.9 \Rightarrow \boxed{N=1}$$

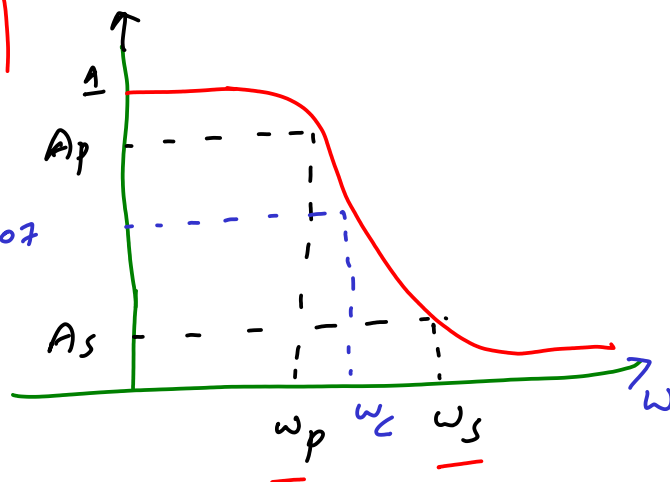
$$N \geq 0.001 \rightarrow \boxed{N=1}$$

$$\checkmark N \geq 3.01 \rightarrow \boxed{N=4}$$

$$N \geq 3.0001 \rightarrow N=4$$

II

$$|H(e^{j\omega})|$$



$$\omega = \Omega T$$

$$\Omega_p = \omega_p / T$$

$$\Omega_s = \omega_s / T$$

→ (11)

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right) \rightarrow \text{BLT}$$

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right)$$

$$\Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right)$$

(11)

find the analog cut off frequency ' Ω_c '

$$\Omega_c = \frac{\Omega_s}{\left[\left(\frac{1}{A_s^2}\right) - 1\right]^{1/2N}}$$

$$(or) \Omega_c = \frac{\Omega_s}{\left[\begin{matrix} 0.1 \alpha_s, db \\ 10 \end{matrix} - 1\right]^{1/2N}}$$

$$\Omega_c = \frac{\Omega_p}{\left[\left(\frac{1}{A_p^2}\right) - 1\right]^{1/2N}}$$

$$(or) \Omega_c = \frac{\Omega_p}{\left[\begin{matrix} 0.1 \alpha_p, db \\ 10 \end{matrix} - 1\right]^{1/2N}}$$

(V) choose the appropriate Normalised $\omega_p, \omega_s, \omega_c, N$
 transfer function based on
 the value of 'N' $\longleftrightarrow$

if $N=1$ $H(s_n) = \frac{1}{s_n + 1}$

$N=2$ $H(s_n) = \frac{1}{s_n^2 + 1.414s_n + 1}$

(vi) find the Actual transfer function

$H(s) = H(s_n)$ at $s_n = \frac{s}{\omega_c}$ (X)

(vi) to design digital butterworth IIR filter

$H(z)$ $\leftarrow$ $H(s)$ $\swarrow$ BLT

(vii) to obtain the frequency response

$H(e^{j\omega}) = H(z)$ at $z = e^{j\omega}$

Note

LPF

$s_n \rightarrow s/\omega_c$ ✓

HPF

$s_n \rightarrow \omega_c/s$

BPF

$s_n \rightarrow \frac{\omega (s^2 + \omega_o^2)}{\omega_o s}$ ✓

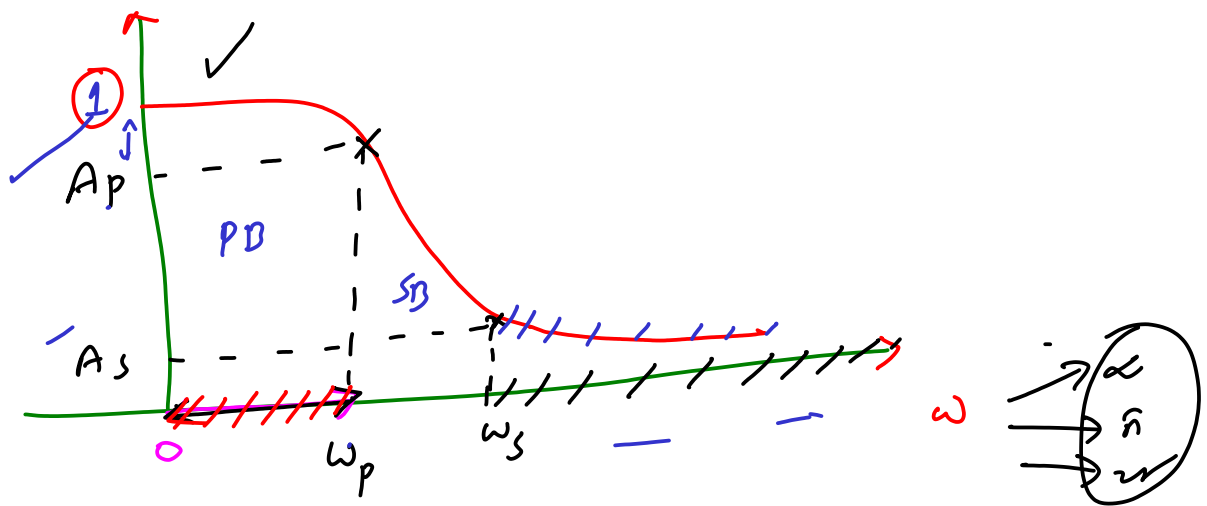
$Q = \frac{\omega_o}{\omega_s - \omega_p}$

$\omega_o = \sqrt{\omega_p \cdot \omega_s}$

BSF

$$s_n \rightarrow \frac{\Omega_0 \cdot s}{\Omega(s^2 + \Omega_0^2)} \quad \checkmark$$

$|H(e^{j\omega})|$



PB $\rightarrow$

$$0 \leq \omega \leq \omega_p$$

$\rightarrow$

$$A_p \leq |H(e^{j\omega})| \leq 1$$

SB $\rightarrow$

$$\omega_s \leq \omega \leq \pi$$

$\rightarrow$

$$|H(e^{j\omega})| \leq A_s$$